Requirement Analysis Phase
Technology Stack (Architecture & Stack)

Date	28 June 2026
Team ID	SWTID-2026-2260
Project Name	Smart Lender
Maximum Marks	4 Marks

Technical Architecture:

The Smart Lender system is designed using a simple and scalable architecture that integrates a user-friendly web interface, application logic, machine learning model, and cloud deployment. The application accepts loan applicant details through a web interface, preprocesses the data, and sends it to the trained XGBoost model for prediction. The model analyzes the applicant's financial and personal information and predicts whether the loan should be approved or rejected. The prediction result is then displayed instantly to the user. The complete application is developed using Python and Flask and is designed for deployment on IBM Cloud to provide secure, reliable, and real-time loan approval prediction.

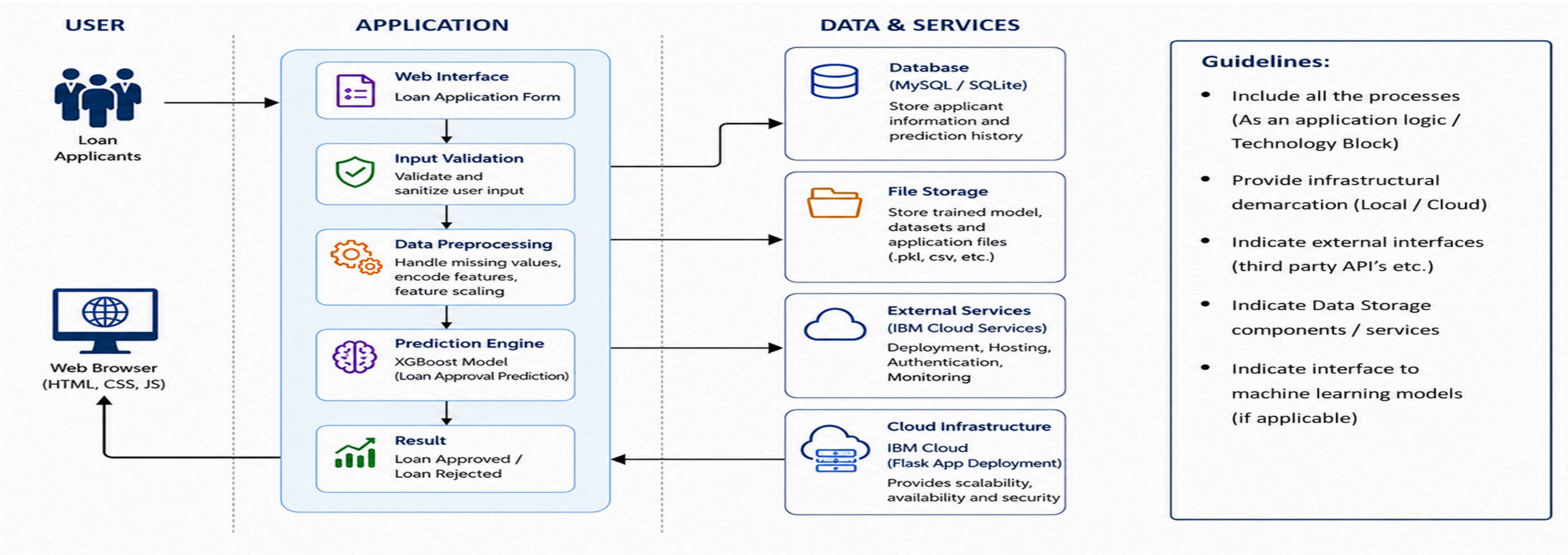

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	Provides a web-based interface for users to enter loan applicant details and view prediction results.	HTML, CSS, Bootstrap, JavaScript
2	Application Logic-1	Receives user input, validates data, and handles application requests.	Python, Flask
3	Application Logic-2	Performs data preprocessing such as handling missing values, encoding categorical features, and feature scaling.	Pandas, NumPy, Scikit-learn
4	Application Logic-3	Loads the trained ML model and predicts loan approval or rejection.	XGBoost, Scikit-learn
5	Database	Stores applicant information, prediction history, and system records.	MySQL / SQLite
6	Cloud Database	Stores application data securely on cloud for scalability.	IBM Cloud Databases
7	File Storage	Stores datasets, trained model (.pkl), and application files.	IBM Cloud Object Storage / Local Storage
8	External API-1	Provides cloud deployment and hosting services for the application.	IBM Cloud Services
9	External API-2	Not required for this project.	N/A

10	Machine Learning Model	Predicts whether the loan application should be approved or rejected based on applicant details.	XGBoost (Final Model), Decision Tree, Random Forest, KNN
11	Infrastructure (Server / Cloud)	Deploys the Flask application and machine learning model for online access.	IBM Cloud, Flask Server

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Open-source frameworks and libraries used for developing the application.	Flask, Scikit-learn, XGBoost, Pandas, NumPy, Bootstrap
2	Security Implementations	Ensures secure data processing through input validation, secure communication, and cloud security.	Flask Validation, HTTPS, IBM Cloud Security
3	Scalable Architecture	Supports modular design and cloud deployment for handling multiple users and future enhancements.	Flask, IBM Cloud
4	Availability	Makes the application accessible anytime through cloud deployment with reliable uptime.	IBM Cloud
5	Performance	Delivers fast and accurate loan approval predictions with optimized model execution and low response	XGBoost, Flask

		time.	